

Survival Strategy Framework

Synthetic Retention Demo

Business Requirements Document (BRD) - Full System Design Artifact

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Document Type	Full System Specification
Project Intent	Portfolio-grade, governed survival-analysis and decision-framework design artifact
Primary Purpose	Controlled time-to-event strategy testing without exposing PII
Implementation	Python 3.12; pandas, NumPy, scikit-learn, lifelines, matplotlib, python-pptx, ReportLab
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----- EXECUTIVE SNAPSHOT -----	
What This Is	A configurable survival strategy framework that combines descriptive persona discovery, regularized CoxPH modeling, risk stratification, governed same-cohort scenario simulation, and multi-audience evidence.
Why It Matters	Binary churn classification answers whether an event may occur. This framework adds when risk emerges, which segments have shorter runway, how controlled assumptions move modeled hazard, and how evidence should be governed.
What This Demonstrates	Applied survival modeling, analytical-system architecture, feature dependency governance, validation design, executive interpretation, and automated evidence generation.
Core Insight	Time-to-event analytics becomes more useful when event definitions, censoring, model behavior, scenario logic, calibration, assumptions, and stakeholder evidence travel together.
Reader Takeaway	This document defines what the system is designed to do, why the architecture is structured this way, how it is governed, what evidence it must produce, and where its use must stop.
The framework is intentionally more than a model. It is a reusable analytical contract and teaching artifact that connects data, statistics, scenarios, validation, lineage, and stakeholder interpretation.	

1. EXECUTIVE SUMMARY

The Survival Strategy Framework establishes a governed, reusable time-to-event analytical environment for synthetic retention and lifecycle strategy testing. Its purpose is to move beyond one-time churn scoring by defining a complete architecture: governed duration and event inputs, independent persona discovery, regularized Cox proportional hazards modeling, risk-tier and target-cohort identification, same-cohort scenario sensitivity, engineered-feature dependency synchronization, calibration and proportional-hazards evidence, and automated stakeholder reporting.

The reference implementation uses deterministic synthetic customer-lifecycle data. It is designed as a portfolio and teaching artifact, not a production retention system. The system demonstrates how a survival model can be embedded inside a broader decision framework where assumptions, controls, evidence, and interpretation remain connected.

Central design thesis: A one-time model produces an answer. A governed framework creates a disciplined way to ask, test, explain, and archive better time-to-event questions.

2. TABLE OF CONTENTS / DOCUMENT NAVIGATION

Section	Title	Purpose
1	Executive Summary	Summarizes the system purpose and design position.
2	Document Navigation	Provides the section map.
3	Business Objective and Strategic Value	Defines the business problem and strategic utility.
4	Design Principles	Establishes reproducibility, separation, explainability, validation, and boundary discipline.
5	Scope	Defines included capabilities, exclusions, and interpretation limits.
6	System Architecture Overview	Describes the five-layer framework and information flow.
7	Data Contract and Synthetic Demonstration Design	Defines identifiers, duration, event, censoring, and input rules.
8	Persona Segmentation Framework	Defines descriptive clustering and persona evidence.
9	CoxPH Model Development Framework	Defines model features, standardization, regularization, fit, and interpretation.
10	Risk Stratification and Target Cohort	Defines scoring, percentile ranking, risk tiers, and same-cohort targeting.
11	Scenario and Dependency-Safe Simulation	Defines control, improvement, stress, scenario changes, bounds, and re-scoring.
12	Validation and Model-Evidence Requirements	Defines CV, calibration, PH, persona quality, controls, and acceptance states.
13	Output Data Model and Artifact Registry	Defines data products, run archive, reports, and consumption patterns.
14	Parameter Governance and Operating Workflow	Defines configuration, change control, and execution workflow.
15	Model Boundaries and Intended Use	Defines approved uses and explicit non-uses.
16	Future Enhancement Roadmap	Defines additive extensions and separate production boundaries.
17	Final Design Statement	Provides the final system-design conclusion.
Appendix A	Default Configuration	Lists primary framework parameters.
Appendix B	Feature and Data Dictionary	Defines model and segmentation inputs.
Appendix C	Requirements Traceability Matrix	Maps business and control requirements to implementation and validation evidence.
Appendix D	Artifact Inventory	Defines the expected run registry.

3. BUSINESS OBJECTIVE AND STRATEGIC VALUE

3.1 Primary Objective

Create a deterministic, configurable, and evidence-producing survival-analysis framework that enables controlled evaluation of time-to-event risk, persona lifecycle differences, model-derived target cohorts, and intervention sensitivities without relying on real customer data.

3.2 Strategic Value

Strategic Capability	Business Value
Runway-based analysis	Estimates when event risk emerges rather than reducing the problem to a single binary score.
Persona lifecycle evidence	Shows how different operational profiles can have different survival trajectories.
Model-derived targeting	Focuses scenario testing on a governed cohort with the shortest modeled runway.
Same-cohort comparison	Isolates scenario assumptions by preserving identical customer identifiers.
Dependency-safe sensitivity	Prevents stale interactions or squared terms from invalidating what-if simulations.
Multi-audience evidence	Generates executive interpretation and technical validation artifacts from the same run.
Teaching transparency	Documents statistical concepts, assumptions, controls, and boundaries directly in code and reports.

3.3 Questions the Framework Is Designed to Support

- When is modeled event risk most likely to emerge?
- Which descriptive personas have the shortest observed lifecycle runway?
- Which records form the model-derived high-risk target cohort?
- How do controlled changes affect mean partial hazard for the same population?
- How does modeled survival at a selected horizon move under improvement and stress assumptions?
- Do scenario changes remain mathematically consistent with engineered interactions and nonlinear terms?
- Can the evidence be reproduced, challenged, and communicated to both executive and technical reviewers?

4. DESIGN PRINCIPLES

Design Principle	Requirement
Deterministic reproducibility	Identical seed, data contract, configuration, and scenarios must produce identical populations and substantive outputs.
Separation of analytical concerns	Persona clustering and CoxPH modeling must remain independent branches unless explicitly designed otherwise.
Time-to-event integrity	Duration, event, censoring, and evaluation horizons must be explicit and validated.
Train-only validation transformations	Cross-validation scalers and models must be fit only on training folds.
Same-cohort scenario attribution	All scenarios must preserve the baseline target IDs.
Dependency synchronization	Interactions and squared terms must be rebuilt after scenario changes.
Bidirectional control testing	The framework must include neutral control and adverse stress evidence, not only favorable scenarios.
Model-evidence hierarchy	Apparent performance must be distinguished from cross-validation and calibration.
Boundary-aware communication	Sensitivity outputs must not be described as causal effects, realized value, or production forecasts.
No cold handoffs	Logic, assumptions, validation, scenario definitions, run lineage, and executive interpretation must remain connected.

5. SCOPE

5.1 Included Capabilities

- Deterministic synthetic time-to-event population generation.
- Unique governed identifiers, positive duration, binary event, and right-censoring design.
- Numeric and Boolean model feature contract with missingness and variance validation.
- K-Means persona segmentation with scaling, descriptive labels, profiles, quality, and stability evidence.
- Regularized CoxPH model with standardized features, coefficients, hazard ratios, confidence intervals, concordance, and PH diagnostics.
- Stratified K-fold validation and out-of-fold survival prediction.
- Horizon-specific calibration and IPCW Brier evidence.
- Model-derived risk ranking, risk tiers, and top-quartile target cohort.
- Control, improvement, combined, and adverse stress scenarios.
- Feature-change bounds, rounding, realized-change audit, and direction checks.
- Interaction and squared-term dependency synchronization.
- Same-cohort partial-hazard and predicted-survival comparison.
- Immutable run registry, CSV/JSON evidence, executive PPTX, and technical PDF.

5.2 Excluded Capabilities

- Real customer data, PII, or proprietary retention strategies.
- Causal treatment-effect estimation.
- Production customer treatment or automated intervention assignment.
- Realized LTV, revenue, ROI, or profitability forecasts.
- External or temporal validation against institution-specific outcomes.
- Full MLOps, deployment, monitoring, or production scoring services.
- Competing-risk or recurrent-event modeling in the current implementation.
- Time-varying coefficient modeling beyond documented PH sensitivity review.

5.3 Interpretation Boundary

The framework answers: “How does modeled time-to-event risk change under this governed assumption set for the same synthetic cohort?” It does not answer: “What intervention will causally retain a real customer?”

6. SYSTEM ARCHITECTURE OVERVIEW



Figure 1. Validated system architecture: input governance, parallel analytics, risk targeting, same-cohort simulation, and evidence archive.

6.1 Architecture Layers

Layer	Purpose	Primary Outputs
1. Survival Input Contract	Defines run scope, identifiers, duration, event/censoring, features, horizons, and engineered dependencies.	Validated input frame; configuration; input fingerprint
2. Parallel Persona + CoxPH Modeling	Produces descriptive personas and an independent regularized survival model.	Persona assignments; profiles; coefficients; hazard ratios; diagnostics
3. Risk Stratification	Scores the population, creates risk tiers, and selects a governed target cohort.	Partial hazard; risk percentile; risk tier; target IDs
4. Governed Same-Cohort Simulation	Applies bounded changes, rebuilds dependencies, re-scores identical IDs, and compares modeled movement.	Scenario scores; hazard reduction; survival uplift; audits
5. Multi-Audience Evidence + Archive	Produces validation evidence, reports, lineage, and reviewer guidance.	PPTX; PDF; CSV/JSON registry; manifest

7. DATA CONTRACT AND SYNTHETIC DEMONSTRATION DESIGN

7.1 Core Record Contract

Field	Type / Rule	Business Meaning
customer_id	Unique string	Governed record identifier used for targeting and scenario matching.
months_observed	Positive numeric months	Observed time to event or censoring.
attrition_event	Binary 0/1	1 = event observed; 0 = right-censored / still active at last observation.
Model features	Numeric / Boolean, finite	Inputs to regularized CoxPH modeling.
Segmentation features	Numeric, scaled	Inputs to K-Means persona discovery.
Engineered terms	Naming: _x_ and _sq	Interactions and squared terms rebuilt during scenarios.

7.2 Synthetic Event and Censoring Design

The demonstration generator creates correlated customer-lifecycle attributes and a proportional-hazards-style event process. Independent censoring is generated separately from event timing. The synthetic process is directional rather than institution-calibrated; it exists to demonstrate framework behavior, not to estimate real retention rates.

Requirement ID	Requirement
SSF-DATA-001	The generator shall be deterministic for a fixed seed.
SSF-DATA-002	The dataset shall contain both events and right-censored records.
SSF-DATA-003	Duration shall be strictly positive and expressed in a configurable time unit.
SSF-DATA-004	Event values shall be restricted to 0 and 1.
SSF-DATA-005	The generator shall preserve mixed-signal profiles rather than deterministic stereotypes.
SSF-DATA-006	The system shall reject invalid identifiers, horizons, features, and scenario references before modeling.

8. PERSONA SEGMENTATION FRAMEWORK

Persona segmentation is a descriptive branch, not an input to the CoxPH model. It uses StandardScaler and K-Means to identify recurring lifecycle profiles and then evaluates observed survival by persona using Kaplan-Meier curves.

Requirement ID	Persona Requirement
SSF-PER-001	Segmentation features shall be standardized before K-Means fitting.
SSF-PER-002	The number of clusters and random state shall be configurable.
SSF-PER-003	Segment IDs shall be reproducible under identical configuration.
SSF-PER-004	Personas shall receive descriptive labels derived from profile evidence.
SSF-PER-005	The framework shall archive centroids, profile summaries, quality metrics, and multi-seed stability.
SSF-PER-006	Personas shall be communicated as descriptive segments, not causal classes or risk grades.

Reference Persona	Design Interpretation
Engaged & Embedded	High onboarding, active use, broader adoption, low service friction, and longer observed runway.
Low-Engagement Risk	Lower onboarding, active use, and adoption, with elevated event risk despite limited service friction.
Service-Friction Risk	Higher support burden, slower resolution, and more incidents, with shortened observed runway.

9. COXPH MODEL DEVELOPMENT FRAMEWORK

9.1 Model Objective

Estimate relative event hazard as a function of governed customer-lifecycle features while preserving explicit distinction between relative hazard, event probability, and predicted survival.

Model Feature	Business Interpretation
onboarding_completion	Degree of completed onboarding; expected protective direction.
monthly_active_days	Engagement frequency; expected protective direction.
product_adoption_count	Breadth of adoption; expected protective direction.
support_tickets_90d	Recent support burden; expected adverse direction.
support_resolution_days	Time to resolve support; expected adverse direction.
service_incidents_90d	Recent service disruptions; expected adverse direction.
tenure_at_start	Existing relationship tenure; expected protective direction.
auto_renew_flag	Renewal commitment indicator; expected protective direction.
enterprise_flag	Enterprise relationship indicator; expected modest protective direction.
monthly_active_days_x_product_adoption_count	Engagement-adoption interaction; expected protective direction.
support_tickets_90d_x_support_resolution_days	Volume-delay interaction; expected adverse direction.
onboarding_completion_sq	Nonlinear onboarding term; interpreted jointly with linear onboarding.

9.2 Model Requirements

Requirement ID	Requirement
SSF-MOD-001	Model features shall be numeric or Boolean and validated before fitting.
SSF-MOD-002	Features shall be standardized when standardization is enabled.
SSF-MOD-003	CoxPH fitting shall use configurable L2/L1 regularization controls.
SSF-MOD-004	The framework shall archive coefficients, hazard ratios, confidence intervals, p-values, concordance, AIC, log likelihood, and fit status.
SSF-MOD-005	Apparent performance shall be labeled separately from cross-validation.
SSF-MOD-006	Proportional-hazards tests shall be archived with raw and multiplicity-adjusted p-values.
SSF-MOD-007	Flagged low-cardinality PH features shall support stratified sensitivity review where appropriate.

10. RISK STRATIFICATION AND TARGET COHORT

The framework converts CoxPH partial-hazard scores into an ordered risk surface. It creates descriptive benchmark, mid-risk, and high-risk tiers and selects the top quantile as the governed scenario cohort.

Requirement ID	Requirement
SSF-RISK-001	Every modeled row shall receive a partial-hazard score and risk percentile.
SSF-RISK-002	The risk-targeting quantile shall be configurable and validated.
SSF-RISK-003	The baseline target cohort shall be nonempty and archived with IDs and profile evidence.
SSF-RISK-004	All scenario comparisons shall use the identical baseline target IDs.
SSF-RISK-005	Risk tiers shall be supported by observed Kaplan-Meier evidence and profile summaries.
SSF-RISK-006	Risk-tier charts shall not be represented as external validation.

The targeting layer identifies where modeled risk is concentrated. It does not authorize customer treatment or claim that a percentile threshold is an approved business policy.

11. SCENARIO AND DEPENDENCY-SAFE SIMULATION

11.1 Scenario Types

Type	Purpose	Expected Evidence
CONTROL	Prove the pipeline introduces no artificial movement when no features change.	Zero hazard and survival movement within tolerance.
IMPROVEMENT	Test favorable sensitivity assumptions on the same target cohort.	Positive hazard reduction and survival uplift.
STRESS	Test adverse bidirectional sensitivity.	Negative hazard reduction and survival uplift.
TECHNICAL	Support portability or diagnostic tests where business direction is unspecified.	Documented technical result.

11.2 Scenario Configuration

Scenario	Type	Expected Direction	Configured Changes
No-Change Control	CONTROL	NEUTRAL	No changes
Onboarding Completion Improvement	IMPROVEMENT	IMPROVED	onboarding_completion add 0.1 [0.0, 1.0]
Support Resolution Improvement	IMPROVEMENT	IMPROVED	support_resolution_days multiply 0.75 [0.25, 20.0]
Product Adoption Expansion	IMPROVEMENT	IMPROVED	product_adoption_count add 1.0 [1.0, 8.0]
Combined Retention Strategy	IMPROVEMENT	IMPROVED	onboarding_completion add 0.1 [0.0, 1.0]; support_resolution_days multiply 0.75 [0.25, 20.0]; product_adoption_count add 1.0 [1.0, 8.0]
Service Friction Stress	STRESS	ADVERSE	support_resolution_days multiply 1.25 [0.25, 20.0]; service_incidents_90d add 1.0 [0.0, 8.0]

11.3 Dependency Synchronization

Scenario changes can invalidate engineered features if dependent terms are not rebuilt. The framework therefore recognizes interaction fields containing _x_ and squared fields ending in _sq, reconstructs them after every scenario change, and archives pre- and post-rebuild discrepancies.

Requirement ID	Requirement
SSF-SCN-001	Scenario definitions shall specify name, type, expected direction, description, and bounded feature changes.
SSF-SCN-002	Scenarios shall modify copies of the baseline target cohort, not the source population.
SSF-SCN-003	Duration, event, identifier, and engineered terms shall not be directly altered by scenarios.
SSF-SCN-004	Configured floors, ceilings, and rounding shall be applied and audited.
SSF-SCN-005	Interaction and squared terms shall be rebuilt before re-scoring.
SSF-SCN-006	The original fitted scaler and CoxPH model shall be reused for scenario scoring.
SSF-SCN-007	The framework shall calculate mean partial-hazard movement and survival uplift at a configured horizon.
SSF-SCN-008	Direction checks shall compare expected and modeled movement.

12. VALIDATION AND MODEL-EVIDENCE REQUIREMENTS

Validation Layer	Required Evidence
Input gatekeeping	ID, duration, event, type, missingness, variance, horizon, scenario, and engineered-term validation.
Internal tests	Deterministic data, dependency rebuilding, invalid event rejection, p-value adjustment, and direction classification.
Discrimination	Apparent concordance plus stratified K-fold out-of-sample concordance.
Calibration	Out-of-fold predicted survival, Kaplan-Meier observed survival, risk-group calibration, and IPCW Brier evidence.
PH assumptions	Raw, Holm, and BH results; residual evidence; sensitivity analysis for eligible review features.
Persona quality	Silhouette, Davies-Bouldin, Calinski-Harabasz, cluster sizes, and multi-seed ARI.
Scenario controls	No-change neutrality, improvement direction, adverse stress, bound audit, same IDs, and finite metrics.
Reproducibility	Exact rerun checks across model, calibration, personas, cohort, scenarios, and curves.
Reporting	Executive deck, technical evidence PDF, and immutable run registry.

12.1 Acceptance States

State	Definition
PASS	No failed or review-status acceptance checks.
PASS_WITH_REVIEW	No failed checks; one or more diagnostics require documented interpretation.
FAIL	At least one structural, computational, reproducibility, direction, probability, or report-creation check failed.

12.2 Calibration-Horizon Support

Calibration horizons should be supported by observed follow-up. The current framework warns when a horizon materially exceeds maximum duration. Future refinement should also flag horizons with inadequate at-risk or censoring support and exclude unsupported endpoints from aggregate prediction-error summaries.

13. OUTPUT DATA MODEL AND ARTIFACT REGISTRY

13.1 Primary Data Outputs

Output	Purpose
input_snapshot.csv	Frozen synthetic source population for the run.
target_cohort.csv	Governed high-risk IDs and profile evidence.
scenario_target_scores.csv	Row-level baseline and scenario partial hazards.
predicted_survival_curves.csv	Baseline and scenario survival curves over the configured timeline.
persona_profiles.csv	Descriptive segment counts, event rates, duration, and feature means.
risk_tier_profiles.csv	Model-derived tier counts, event rates, duration, hazard, and profile means.

13.2 Validation and Governance Outputs

Output Family	Examples
Configuration and lineage	framework_config.json, run_metadata.json, scenario_definitions.json, artifact_manifest.json
Model evidence	model_summary.csv, model_fit_metadata.csv, model_scaler_parameters.csv, baseline survival/hazard
Validation evidence	cross-validation, calibration, PH tests, residuals, sensitivity, persona quality, reproducibility
Scenario evidence	scenario results, change audit, dependency audit, target scores, curves
Stakeholder evidence	Survival_Strategy_Deck.pptx, Technical_Model_Evidence.pdf, executive_narrative.txt

13.3 Output Consumption

Valid Consumption	Invalid Consumption
Portfolio demonstration and teaching	Real customer treatment
Synthetic scenario sensitivity	Causal impact claim
Relative-risk interpretation	Event probability interpretation of partial hazard
Validation and architecture review	Production model approval
Executive and technical communication	Realized ROI, revenue, or LTV claim

14. PARAMETER GOVERNANCE AND OPERATING WORKFLOW

Parameter Category	Examples	Governance Expectation
Run identity	Project name, output root, random state	Recorded in configuration and run metadata.
Time-to-event	Duration, event, time unit, horizons	Validated for positivity, binary values, and support.
Model	Features, penalizer, L1 ratio, standardization	Documented and held stable within scenario comparisons.
Segmentation	Features, cluster count, random state	Separate from CoxPH model features and validated for stability.
Targeting	Risk quantile	Documented and applied once to establish baseline IDs.
Scenario	Name, type, direction, feature changes, bounds	Audited and applied only to the baseline target cohort.
Validation	CV folds, calibration groups, PH alpha, thresholds	Reported in run evidence and acceptance checks.
Reporting	CI toggle, timeline points, snapshot flag	Applied consistently and recorded.

14.1 Recommended User Workflow

1. Define the time-to-event business question, event, censoring rule, and evaluation horizon.
2. Prepare a governed numeric/Boolean input contract and engineered-feature naming convention.
3. Configure model, persona, targeting, validation, and scenario controls.
4. Run input validation and internal self-tests before model fitting.
5. Fit independent persona and CoxPH branches.
6. Review coefficients, cross-validation, calibration, and PH diagnostics.
7. Establish the baseline target cohort once.
8. Apply control, improvement, and stress scenarios to the same IDs.
9. Review scenario-change and dependency audits.
10. Review executive and technical evidence, document boundaries, and archive the run.

15. MODEL BOUNDARIES, INTENDED USE, AND GOVERNANCE GUARDRAILS

Boundary Area	System Position
Data	Synthetic demonstration data; no real customer records or PII.
Partial hazard	Relative risk measure, not event probability.
Predicted survival	Fitted-model output requiring domain and external validation before production use.
Scenarios	Modeled sensitivity under controlled assumptions, not causal treatment effect.
Value	No realized revenue, LTV, ROI, or defended-value claim.
Personas	Descriptive segments, not causal classes or approved customer treatments.
Model validation	Portfolio evidence does not replace formal independent regulatory validation.
Deployment	No API, MLOps, production monitoring, or operational decision authority.

15.1 Boundary-Aware Communication

Avoid Saying	Prefer Saying
The model predicts which customers will churn.	The framework estimates relative time-to-event risk in synthetic demonstration data.
This intervention improves retention by 18.8%.	The configured scenario increased modeled 12-month survival by 18.8 percentage points for the same synthetic target cohort.
The framework defended \$X of LTV.	The framework does not claim realized or defended financial value.
These personas are true customer types.	These are stable descriptive segments in the demonstration population.
The top quartile should receive treatment.	The top quartile is a model-derived demonstration cohort for sensitivity testing.

16. FUTURE ENHANCEMENT ROADMAP

Tier	Enhancement	Design Implication
1 - Near-term	At-risk support validation for calibration horizons	Flag or omit horizons with insufficient support.
1 - Near-term	Report chart label refinement	Improve negative-value label placement.
1 - Near-term	Config-file ingestion	Allow YAML/JSON configuration without code edits.
2 - Analytical	External / temporal validation	Test transportability and stability on governed real or benchmark data.
2 - Analytical	Time-varying covariates or effects	Address dynamic features and non-proportional effects.
2 - Analytical	Competing risks / recurrent events	Extend beyond a single terminal event.
2 - Strategy	Cost, feasibility, and capacity layer	Convert scenario comparison into a governed strategy frontier.
3 - Production	Deployment, monitoring, and model-risk lifecycle	Requires separate architecture, approvals, controls, and validation.
3 - Causal	Experimental or quasi-experimental intervention analysis	Required before treatment-effect claims.

17. FINAL DESIGN STATEMENT

The Survival Strategy Framework is designed as a governed analytical system, not a one-time notebook or isolated CoxPH model. It connects a survival-style data contract, independent persona and model branches, cross-validated model evidence, risk targeting, dependency-safe same-cohort simulation, bidirectional scenario controls, immutable run lineage, and stakeholder reporting.

The framework is complete for its current portfolio purpose: controlled synthetic time-to-event strategy simulation, educational interpretation, and architecture demonstration. It is intentionally bounded before causal, financial, operational, or production claims.

FINAL DESIGN POSITION: A reusable, governed, explainable, scenario-capable survival-analysis framework suitable for public portfolio demonstration and controlled synthetic strategy testing.

APPENDIX A. DEFAULT CONFIGURATION

Parameter	Default
duration_col	months_observed
event_col	attrition_event
id_col	customer_id
time_unit_label	months
n_clusters	3
risk_quantile	0.75
penalizer	0.1
l1_ratio	0.0
random_state	42
cross_validation_folds	5
calibration_horizons	[6.0, 12.0, 18.0, 24.0]
calibration_groups	10
evaluation_horizon	12.0
ph_test_alpha	0.05
timeline_points	121
verify_reproducibility	True

APPENDIX B. FEATURE AND DATA DICTIONARY

Field	Definition	Use
onboarding_completion	Degree of completed onboarding; expected protective direction.	Model + segmentation
monthly_active_days	Engagement frequency; expected protective direction.	Model + segmentation
product_adoption_count	Breadth of adoption; expected protective direction.	Model + segmentation
support_tickets_90d	Recent support burden; expected adverse direction.	Model + segmentation
support_resolution_days	Time to resolve support; expected adverse direction.	Model + segmentation
service_incidents_90d	Recent service disruptions; expected adverse direction.	Model + segmentation
tenure_at_start	Existing relationship tenure; expected protective direction.	Model + segmentation
auto_renew_flag	Renewal commitment indicator; expected protective direction.	Model feature
enterprise_flag	Enterprise relationship indicator; expected modest protective direction.	Model feature
monthly_active_days_x_product_adoption_count	Engagement-adoption interaction; expected protective direction.	Model feature
support_tickets_90d_x_support_resolution_days	Volume-delay interaction; expected adverse direction.	Model feature
onboarding_completion_sq	Nonlinear onboarding term; interpreted jointly with linear onboarding.	Model feature

APPENDIX C. REQUIREMENTS TRACEABILITY MATRIX

Requirement	Business / Control Need	Implementation Anchor	Validation Evidence
SSF-DATA-001	Deterministic synthetic population	Generator + self-test	Synthetic data reproducibility PASS
SSF-DATA-002	Events and censoring present	Input validation	39.52% events / 60.48% censored
SSF-PER-003	Persona reproducibility	Persona stability + rerun	Mean ARI 0.992; rerun exact
SSF-MOD-004	Archive full CoxPH evidence	Model summary / metadata	Coefficients, HRs, CIs, CV, fit status
SSF-MOD-005	Separate apparent and CV performance	Model metadata / deck	Apparent 0.827; CV mean 0.827
SSF-MOD-006	PH evidence and multiplicity adjustment	PH test / residual / sensitivity	Raw review; adjusted pass; sensitivity stable
SSF-RISK-004	Same target IDs across scenarios	Target cohort + scenario scores	1,875 identical IDs
SSF-SCN-004	Audit bounds and realized changes	Scenario change audit	All scenarios archived
SSF-SCN-005	Rebuild dependent features	Dependency audit	All post-rebuild differences zero
SSF-SCN-008	Direction checks	Scenario results / acceptance checks	All control, improvement, stress checks pass
SSF-VAL-001	Cross-validation	Five-fold CV	All folds pass
SSF-VAL-002	Calibration	OOF calibration / Brier	Status PASS; max group error 7.08%
SSF-GOV-001	Immutable run archive	Artifact manifest	42 named artifacts
SSF-GOV-002	Boundary-aware reports	PPTX / PDF / narrative	Synthetic and non-causal caveats present

APPENDIX D. EXPECTED RUN ARTIFACT INVENTORY

Category	Artifact
Stakeholder Output	Survival_Strategy_Deck.pptx
Stakeholder Output	Technical_Model_Evidence.pdf
Run Evidence	acceptance_checks.csv
Run Evidence	calibration_by_risk_group.csv
Run Evidence	calibration_metrics.csv
Chart	charts/baseline_vs_scenario_survival.png
Chart	charts/calibration_at_horizons.png
Chart	charts/persona_kaplan_meier.png
Chart	charts/ph_assumption_diagnostics.png
Chart	charts/risk_tier_kaplan_meier.png
Chart	charts/scenario_hazard_reduction.png
Run Evidence	cox_baseline_cumulative_hazard.csv
Run Evidence	cox_baseline_survival.csv
Run Evidence	cross_validated_predictions.csv
Run Evidence	cross_validation_results.csv
Run Evidence	dependency_audit.csv
Run Evidence	executive_narrative.txt
Run Evidence	framework_config.json
Run Evidence	input_snapshot.csv
Run Evidence	input_validation.csv
Run Evidence	model_fit_metadata.csv
Run Evidence	model_fit_warnings.csv
Run Evidence	model_scaler_parameters.csv
Run Evidence	model_summary.csv
Run Evidence	persona_centroids_raw_scale.csv
Run Evidence	persona_centroids_standardized.csv
Run Evidence	persona_profiles.csv
Run Evidence	persona_quality_metrics.csv
Run Evidence	persona_scaler_parameters.csv
Run Evidence	persona_stability_results.csv
Run Evidence	ph_sensitivity_results.csv
Run Evidence	predicted_survival_curves.csv

Category	Artifact
Run Evidence	proportional_hazards_test.csv
Run Evidence	reproducibility_checks.csv
Run Evidence	risk_tier_profiles.csv
Run Evidence	run_metadata.json
Run Evidence	scaled_schoenfeld_residuals.csv
Run Evidence	scenario_change_audit.csv
Run Evidence	scenario_definitions.json
Run Evidence	scenario_results.csv
Run Evidence	scenario_target_scores.csv
Run Evidence	target_cohort.csv